

Micro-optical Research Scientist/Engineer at Biomedical Diagnostics Startup

We are a Boston-area biomedical diagnostics startup company seeking a highly skilled optical research scientist or engineer with specific expertise in accurate capture and measurement micron-scale optical phenomena via imaging systems. The ideal candidate will have demonstrated experience building and customizing inexpensive robust microscope systems.

Required Qualifications

- Strong training (ideally MS, MEng, or PhD) in optics, physics, or a related field with a strong focus on micro-optics, optical metrology, and light-matter interactions
- Extensive knowledge of optical physics, including wave optics, diffraction theory, and their application to spherical/monodisperse and non-spherical/polydisperse microparticles
- Proficiency in analyzing images singly and in bulk with Python and the relevant scientific computing libraries (e.g., NumPy, SciPy, scikit-learn) and programs (Fiji)
- Demonstrated experience in microscope construction, assembly, and/or development
- Experience in developing test methods for micro-optical systems
- Proficiency in commercial and open-source optical design and simulation software (e.g. Zemax OpticStudio, COSMOL Multiphysics, LightTools, Ray Optics, PyOpTools)
- Experience measuring/reconstructing point-spread functions (PSFs), super-resolution microscopy techniques, and image processing algorithms
- Strong record-keeping and written and oral communication skills
- Exceptional analytical, problem-solving, and communications skills with meticulous attention to detail
- Professionalism, a sense of mission, and a desire to work with others who feel the same in a fast-paced, collaborative research environment

Preferred Qualifications

- Experience with deep learning for image processing, generation, and cleanup
- Exceptional publication record (arXiv) and/or publication in reputable optics journals/conference proceedings (e.g. IEEE, ACS)

Particulars

- Company headquarters: Beverly, MA; on-site work required
- Competitive salary and equity compensation package
- Comprehensive health benefits and flexible paid time off

Why Join Us?

With the above qualifications, you could work anywhere. But ask yourself: do you want to be just a cog in someone else's machine? Punching a clock for a paycheck? Working on something sad, boring, or meaningless? Or do you want to be a mission-critical part of building things to save lives? If the answer to that last question is "yes," please send a short (~250-word) cover letter and resume/CV, including any links to your GitHub or other relevant work, to

careers@astradx.com.

Note: Your cover letter should:

- Explain what it is about us, and/or about your background, that makes you interested in us specifically (~100 words)
- Mention one concept, idea, invention, or discovery that you've come across in the past year or so that you've found **really cool**, and describe **why** (~100 words). It can be related to this area, your own work, or something else